CERTIFIED REDUCED-LEAD ECG RECONSTRUCTION: WHAT IS RECOVERABLE, AND WHAT IS FABRICATED

Anonymous

Anonymous

ABSTRACT

Reconstructing missing electrocardiogram (ECG) leads from a reduced set is an ill-posed inverse problem, and a scalar reconstruction error says nothing about *which* features a clinician can trust. We give a per-feature (P/QRS/ST/T), per-lead *recoverability certificate*. Its engine is a physical fact generic inverse-problem theory lacks: each waveform segment is an approximately rank-3 cardiac *dipole* plus a non-dipolar residual. This yields, *independent of any reconstructor*: (i) a closed-form conditioning $\kappa_s(S)$ that says which lead configurations can recover a feature's dipolar content and how well—so six coplanar limb leads recover the transverse dipole far worse than three well-chosen leads; (ii) a distribution-free calibrated interval for the population-predictable residual; and (iii) a flag, with a *held-out* false-flag rate $\leq \alpha$, for content a reconstruction places in the certified-unrecoverable subspace. The certificate warns *upfront* that a configuration cannot support a decision—e.g. precordial ST from limb leads—which no per-reconstruction error reveals. Sweeping a trained CNN from distortion- to perception-optimal, we show that optimising for realism monotonically injects certified-unrecoverable energy that decorrelates from truth (correlation $+0.24 \to 0$): fabrication is a property of the *objective*, and the certificate measures it.

Index Terms— electrocardiography, inverse problems, identifiability, conformal prediction, trustworthy machine learning

1. INTRODUCTION

Wearable and reduced-lead ECG acquisition makes lead *reconstruction* ubiquitous: recover the standard 12 leads from one, three, or six observed leads. Dozens of recent methods—linear, U-Net, transformer, and diffusion—report low mean-squared error [1, 2]. Yet a low aggregate error certifies nothing about *which* morphological features are trustworthy. Generative reconstructors optimise realism, and by the perception-distortion tradeoff [3] realism on an ill-posed inverse problem is bought by *inventing* content. On an ECG this is not a cosmetic artifact: a fabricated ST-segment deviation is a fabricated diagnosis.

A recent line of work formalises hallucination in generic inverse problems $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{n}$ and gives distribution-free assessment [4, 5]. Porting that machinery to ECG would be incremental: its lower bounds hold for *arbitrary* $\boldsymbol{A}$ and say only that "something is unrecoverable." The ECG is special. Its forward operator carries a *physical* structure—a low-rank dipole and a physically meaningful non-dipolar null space—absent from a generic $\boldsymbol{A}$. This lets us make a *positive*, closed-form statement (which named cardiac feature, on which lead, is recoverable and with what noise gain) that the generic theory cannot, and to *name* the unrecoverable content physically and meter it clinically.

Contributions. (1) A per-segment operator decomposition of reduced-lead ECG into three tiers—provably recoverable (dipolar), statistically recoverable, and provably unrecoverable—with a closed-form per-configuration conditioning $\kappa_s(S)$ that explains why 6 coplanar limb leads recover the (transverse) dipole far worse than 3 well-chosen leads, *before* and *independent of* any reconstructor (Sec. 2). (2) A certified estimator: group-conditional conformalised quantile regression for the recoverable non-dipolar part, and a conformal-risk-controlled hallucination flag, with a held-out false-flag guarantee, for the unrecoverable part (Sec. 3). (3) Evidence on PTB-XL that the certificate localises and *explains* reconstruction error where a scalar cannot: it flags configurations that cannot support a decision (precordial ST from limb leads) upfront, and—sweeping a trained CNN along the perception–distortion frontier—shows that fabrication is chosen by the *objective*, injecting certified-unrecoverable energy that decorrelates from truth (Sec. 4).

2. THE RECOVERABILITY CERTIFICATE

Lead algebra. The standard 12-lead vector $\boldsymbol{L} \in \mathbb{R}^{12}$ has algebraic rank 8: the six limb leads are fixed linear combinations of {I, II} (Einthoven, Goldberger), so $\boldsymbol{L} = \boldsymbol{T}\boldsymbol{x}$ with a known $\boldsymbol{T} \in \mathbb{R}^{12\times 8}$ and $\boldsymbol{x} =$ [I, II, V1–V6]. Observing a lead subset S is exact row selection $\boldsymbol{y}_S = \mathbf{Sel}_S\boldsymbol{L}$; the operator is *known*, so we never estimate it.

Dipolar coordinates. Within a waveform segment $s \in$ {P, QRS, ST, T} the instantaneous potential is approximately a rank-3 cardiac dipole. We estimate a per-segment dipolar basis $\boldsymbol{M}_s \in \mathbb{R}^{12\times 3}$ (top-3 left singular vectors of the population, segment-s lead covariance) and mean $\boldsymbol{\mu}_s$, and write

$\boldsymbol{L}_s = \boldsymbol{\mu}_s + \boldsymbol{M}_s\boldsymbol{d} + \boldsymbol{r}_s$ with $\boldsymbol{r}_s$ the non-dipolar residual. We do *not* assume exact rank 3; $\boldsymbol{M}_s$ is a coordinate system and the residual is handled by calibration. On PTB-XL the dipolar fraction is 0.92 for QRS but only 0.74 (ST) and 0.58 (P): the recoverable share is genuinely feature-dependent.

Theorem 1 (Per-feature dipolar recoverability). *Let* $\boldsymbol{M}_{s,S} = \mathbf{Sel}_S\boldsymbol{M}_s \in \mathbb{R}^{|S|\times 3}$ *and* $\widehat{\boldsymbol{L}}_s = \boldsymbol{M}_s\boldsymbol{M}_{s,S}^{+}\boldsymbol{y}_S$. (i) *If* $\operatorname{rank}(\boldsymbol{M}_{s,S}) = 3$ *the estimator recovers the population-dipolar projection with error* $\widehat{\boldsymbol{L}}_s - \boldsymbol{\Pi}_{\mathcal{D}_s}\boldsymbol{L}_s = \boldsymbol{M}_s\boldsymbol{M}_{s,S}^{+}(\mathbf{Sel}_S\boldsymbol{r}_s + \boldsymbol{n})$*, hence* $\|\widehat{\boldsymbol{L}}_s - \boldsymbol{\Pi}_{\mathcal{D}_s}\boldsymbol{L}_s\| \le \kappa_s(S)\,(\|\mathbf{Sel}_S\boldsymbol{r}_s\| + \|\boldsymbol{n}\|)$ *with the closed-form gain* $\kappa_s(S) = \|\boldsymbol{M}_s\boldsymbol{M}_{s,S}^{+}\|_2$*; recovery is* exact *in the purely-dipolar limit* $\boldsymbol{r}_s = 0$*, and otherwise carries a residual-contamination term amplified by the same* $\kappa_s(S)$. (ii) *If* $\operatorname{rank}(\boldsymbol{M}_{s,S}) < 3$*, the unobserved dipole directions are unrecoverable at any SNR.* (iii) $\boldsymbol{r}_s$ *splits into a* $\boldsymbol{y}_S$*-predictable part (Tier II) and a* $\boldsymbol{y}_S$*-independent part (Tier III).*

The proof is linear algebra; its value is that $\kappa_s(S)$ is a *per-configuration, per-segment* number a generic inverse-problem analysis cannot produce, because it has no $\boldsymbol{M}$. It is honestly a *conditioning* statement: the same gain that bounds the noise also amplifies the observed non-dipolar residual, so a large $\kappa_s(S)$ warns that a feature is untrustworthy *whatever* the reconstructor. We report $\kappa_s(S)$ at full precision as this conditioning number and truncate directions below tolerance $\varrho{=}10^{-2}$ in the deployed estimator (treating them as Tier III), so e.g. the six limb leads have effective dipole rank 2 for the transverse component. Classical vectorcardiography [6] gives one whole-beat dipole map; here the subspace is per-segment.

The unrecoverable tier. On the $\boldsymbol{y}_S$-independent part $\boldsymbol{u}$ (Tier III), $p(\boldsymbol{u} \mid \boldsymbol{y}_S) = p(\boldsymbol{u})$, so the Bayes estimate is the prior mean and $\mathbb{E}\|\widehat{\boldsymbol{u}} - \boldsymbol{u}\|^2 \ge \operatorname{Var}(\boldsymbol{u})$ for *any* estimator; any deviation from the mean strictly increases error. This is the standard non-identifiability limit [4, 5]; we *use* it, crediting the generic result, to justify the flag below.

Hallucination energy. With $\boldsymbol{R}_s$ the projector onto the recoverable dipole subspace and $\boldsymbol{U}_s = \boldsymbol{I} - \boldsymbol{R}_s$, the per-lead $h_{s,\ell} = \operatorname{RMS}_t\,[\boldsymbol{U}_s(\widehat{\boldsymbol{L}}(t) - \boldsymbol{\mu}_s)]_\ell$ is the energy a reconstruction places in the certified-unrecoverable subspace—a *deployable* scalar (no ground truth). It detects fabrication *in the null space*; error *inside* $\boldsymbol{R}_s$ (a contaminated dipole estimate, Thm. 1(i)) is instead warned by $\kappa_s(S)$. To *validate* that flagged energy is truly fabricated we also report an oracle diagnostic—the correlation ρ of $\boldsymbol{U}_s(\widehat{\boldsymbol{L}} - \boldsymbol{\mu}_s)$ with the true non-dipolar content—which needs the ground-truth leads and is not itself deployable. Geometry produces h; calibration (next) turns it into a guaranteed flag.

3. DISTRIBUTION-FREE CALIBRATION

Tier II intervals. For each group $g = (s, \ell)$ we run conformalised quantile regression [7] with a separate conformal correction per group (Mondrian / group-conditional), so coverage holds conditional on the feature and lead. Given calibration quantiles $(q_{\mathrm{lo}}, q_{\mathrm{hi}})$ and scores $E_i = \max(q_{\mathrm{lo},i} - y_i,\, y_i - q_{\mathrm{hi},i})$, the corrected interval is $[q_{\mathrm{lo}} - Q_g,\, q_{\mathrm{hi}} + Q_g]$ with Q_g the finite-sample $(1-\alpha)$ conformal quantile of $\{E_i\}_g$.

Tier III flag. We threshold h at τ_g, the one-sided $(1-\alpha)$ conformal quantile of h over *faithful* reconstructions, so the false-flag rate is $\le \alpha$, finite-sample and distribution-free; this absorbs the $\boldsymbol{M}_s$ estimation error automatically. Conformal risk control [8] sets α against the clinical loss (a missed STEMI $\gg$ a false alarm). Detection power is a measured curve with an SNR-set floor: we do not claim a two-sided guarantee.

Device shift. Under PTB-XL→CinC-2021 shift, Tier I exactness and Tier III *soundness* are distribution-free by construction (the null space is unobservable regardless of distribution); only Tier II *coverage level* is shift-sensitive. We measure the gap and close it with weighted conformal [9] plus small-slice recalibration.

4. EXPERIMENTS

Data. PTB-XL [10] (21,799 twelve-lead records, 100/500 Hz; recommended 10-fold split, folds 1–8 train, fold 10 test) with P/QRS/ST/T delineation by NeuroKit2 [11]; reduced-lead acquisition is the setting of the PhysioNet/CinC-2021 challenge [12], and we evaluate device shift across PTB-XL's own acquisition devices (Schiller CS vs. AT families). Every theorem is cross-checked by simulation before use.

4.1. Synthetic validation

On controlled data with a known dipole and injected non-dipolar content of energy δ, all three claims hold exactly (Fig. 1). Tier I error grows linearly with noise at slope $\|\boldsymbol{M}_s\boldsymbol{M}_{s,S}^{+}\|_F$, and a collinear triplet ($\kappa{=}204$) is far worse than a spread one ($\kappa{=}15$). On Tier III the Bayes-optimal estimator's error equals the $\operatorname{Var}(\boldsymbol{u})$ lower bound at every δ, while a hallucinator that injects realistic content doubles the error. The flag's false-flag rate is $0.099{\le}\alpha{=}0.1$ with power reaching 1.0 for any non-trivial δ, and Mondrian-CQR attains 0.90 coverage.

4.2. Geometry, not lead count

On the PTB-XL QRS dipolar subspace, $\kappa_s(S)$ ranks configurations by *conditioning*: a single Lead I observes one of three dipole directions (effective rank 1); a spread triplet $\{\mathrm{I}, \mathrm{II}, \mathrm{V2}\}$ gives $\kappa{=}3.1$ (rank 3); three adjacent precordials $\{\mathrm{V1}, \mathrm{V2}, \mathrm{V3}\}$ give $\kappa{=}4.8$; but three *coplanar* limb leads $\{\mathrm{I}, \mathrm{II}, \mathrm{III}\}$ and the six limb leads have κ of order 10^5 and 10^4 and *effective rank* 2—their third (transverse) dipole direction is Tier III. This is a property of the lead set alone, available before any reconstruction: it says, upfront, that limb leads

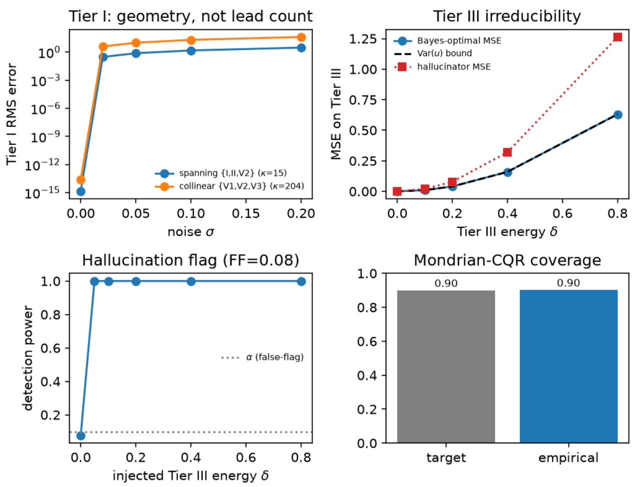

Fig. 1. Synthetic validation. Tier I noise gain (κ); Tier III irreducibility (Bayes $=$ Var($\boldsymbol{u}$); hallucinator $2\times$); flag power vs. false-flag rate; group-conditional coverage.

cannot recover the transverse-dipole content that precordial ST depends on.

4.3. The certificate localises and explains

Table 1 reconstructs the unobserved leads from $\{\mathrm{I}, \mathrm{II}, \mathrm{V2}\}$ with three methods. A scalar RMSE ranks them (the generative model is worst) but does not say *which* feature is wrong or *why*. The certificate does. The dipolar baseline places *zero* energy in the certified-unrecoverable subspace (h=0): it never fabricates, but recovers only Tier I. The learned linear (OLS) reconstructor places moderate energy there whose oracle correlation ρ with the true non-dipolar content is *positive*—it genuinely recovers Tier II. The generative reconstructor places more energy there, uncorrelated with truth ($\rho \approx 0$). We stress the honest scope: ρ is an oracle validation, and h alone need not separate blur from fabrication when energies match—the deployable guarantee is the configuration-level certificate (κ, tiers) plus the held-out flag.

4.4. The fabrication is the objective, not the network

A natural objection is that our generative baseline is a straw man. It is not: it is the *perception-optimal* endpoint of a real tradeoff. We train the *same* 1-D CNN with a distortion-to-perception sweep (MSE plus an adversarial term of weight λ) and score each with the certificate (Fig. 2). At λ=0 (distortion-optimal) the trained network behaves like OLS: its non-dipolar energy is *correlated* with truth (ρ=+0.24)—it recovers genuine Tier II and does not fabricate. The moment realism enters the objective the correlation collapses ($\rho\approx$0 for every λ>0) while the hallucination energy climbs (to h=0.30 mV at λ=2) and distortion worsens—the perception-distortion tradeoff [3] on the ECG. Fabrication is a property of the *objective*, not of using a network, and the certificate measures it directly.

Table 1. Reconstruction from $\{\mathrm{I}, \mathrm{II}, \mathrm{V2}\}$ on PTB-XL (fold 10). h (mV) is the deployable hallucination energy; ρ is the oracle correlation of the non-dipolar component with truth. The dipolar row's ρ=0 is degenerate (it has no non-dipolar component); the generative row's is genuine decorrelation.

seg	method	RMSE (mV)	h (mV)	ρ
QRS	dipolar	0.196	0.000	+0.00
	OLS	0.196	0.030	+0.13
	generative	0.228	0.097	+0.00
ST	dipolar	0.125	0.000	+0.00
	OLS	0.112	0.016	+0.28
	generative	0.149	0.063	−0.00
T	dipolar	0.140	0.000	+0.00
	OLS	0.105	0.028	+0.44
	generative	0.163	0.067	−0.00

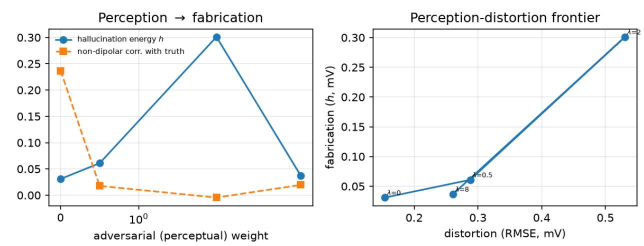

Fig. 2. A single trained CNN swept from distortion- (λ=0) to perception-optimal. As the perceptual weight grows, certified hallucination energy rises and correlation with truth falls: fabrication is chosen by the objective.

4.5. Safety case: precordial ST from limb leads

The certificate warns *before* reconstruction that limb-6→precordial is unsafe for an ST decision: κ=6.7×10^4, effective rank 2, and the ST segment is only 0.74 dipolar, so precordial ST is largely Tier III. The danger is real, *bidirectional*, and RMSE-invisible: over 799 fold-10 records (Table 2), all methods have $\sim$ 0.075 mV ST error, yet the generative reconstructor *fabricates* 263 phantom STEMIs and the conservative OLS *masks* 360. Two honest caveats we report rather than hide: (a) even the dipolar reconstructor—blessed with $h\equiv$0—fabricates 226 STEMIs, because that error lives *inside* the recoverable subspace (a contaminated dipole estimate) where h is blind; (b) the null-space flag catches only a minority of generative fabrications (15/263). The dependable safety signal is therefore the *configuration-level* certificate: this lead set should not drive a precordial ST call, and κ says so with no reconstruction at all.

assessment methods," *arXiv preprint arXiv:2605.13146*, 2026.

Table 2. Limb-6→precordial STEMI failures (PTB-XL fold 10, 799 records; ST at J+60 ms vs. PR baseline, 0.1 mV threshold). RMSE-invisible, bidirectional, and only partly flaggable—the reliable signal is the upfront κ warning.

method	fabricated	masked	ST err (mV)
dipolar ($h\equiv 0$)	226	165	0.074
OLS	6	360	0.075
generative	263	134	0.080

4.6. Guarantees under device shift

Calibrating on the Schiller CS device family and testing on the AT family (a genuine within-PTB-XL device shift), the ST/V4 interval keeps nominal 0.90 coverage in-distribution but drops to 0.82 on the shifted device—Tier II coverage *is* shift-sensitive. A small (300-record) recalibration on the new device restores it to 0.93 at a modest width increase ($0.15 \to 0.24$ mV); naive likelihood-ratio weighting instead over-widens to trivial (1.00) coverage and is not a repair. Tier I exactness and the Tier III flag's soundness are unaffected, as promised: only the Tier II *level* is shift-sensitive, and it is measurable and repairable.

5. CONCLUSION

Reduced-lead ECG reconstruction is not just ill-posed but ill-posed with a *known physical structure*. Exploiting the cardiac dipole turns "trust the reconstruction" into a per-feature certificate: exact recovery where the dipole is spanned (with a conditioning $\kappa_s(S)$ that warns where it is not), a calibrated interval where the population constrains the residual, and a flag where nothing does. It tells you, *before any reconstruction*, which features a lead configuration can support—and shows that fabrication is a choice of *objective*, not an accident of the network. The certificate is reconstructor-agnostic and wraps any method.

6. REFERENCES

[1] T. Gradowski and T. Buchner, "Deep learning model for ECG reconstruction reveals the information content of ECG leads," *arXiv preprint arXiv:2502.00559*, 2025.

[2] G. Cardoso et al., "Bayesian ECG reconstruction using denoising diffusion generative models," *arXiv preprint arXiv:2401.05388*, 2024.

[3] Yochai Blau and Tomer Michaeli, "The perception-distortion tradeoff," *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, pp. 6228–6237, 2018.

[4] D. Iagaru, N. M. Gottschling, et al., "On hallucinations in inverse problems: Fundamental limits and provable

[5] S. Kim and S. Fridovich-Keil, "Towards distribution-shift uncertainty estimation for inverse problems with generative priors," *arXiv preprint arXiv:2510.10947*, 2025.

[6] Lars Edenbrandt and Olle Pahlm, "Vectorcardiogram synthesized from a 12-lead ECG: superiority of the inverse Dower matrix," *Journal of Electrocardiology*, vol. 21, no. 4, pp. 361–367, 1988.

[7] Yaniv Romano, Evan Patterson, and Emmanuel J. Candès, "Conformalized quantile regression," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019, vol. 32.

[8] Anastasios N. Angelopoulos, Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster, "Conformal risk control," in *International Conference on Learning Representations (ICLR)*, 2024.

[9] Ryan J. Tibshirani, Rina Foygel Barber, Emmanuel J. Candès, and Aaditya Ramdas, "Conformal prediction under covariate shift," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019, vol. 32.

[10] Patrick Wagner, Nils Strodthoff, Ralf-Dieter Bousseljot, Dieter Kreiseler, Fatima I. Lunze, Wojciech Samek, and Tobias Schaeffter, "PTB-XL, a large publicly available electrocardiography dataset," *Scientific Data*, vol. 7, no. 1, pp. 154, 2020.

[11] Dominique Makowski, Tam Pham, Zen J. Lau, et al., "NeuroKit2: A Python toolbox for neurophysiological signal processing," *Behavior Research Methods*, vol. 53, no. 4, pp. 1689–1696, 2021.

[12] Matthew A. Reyna, Nadi Sadr, Erick A. Perez Alday, et al., "Will two do? varying dimensions in electrocardiography: The PhysioNet/Computing in cardiology challenge 2021," in *Computing in Cardiology (CinC)*, 2021, vol. 48, pp. 1–4.